# Force Diagrams

Exam code: 9MA0

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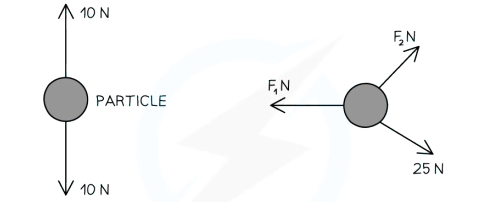
## Force diagrams

### Why do we need force diagrams?

- **Force diagrams** are used to help understand a given scenario and show which **forces** are acting on which **particles** and in which **direction** they are acting
- In diagrams an **arrow** is used to represent a **force** acting on a **particle** which shows the **direction** in which the force is acting
- The **magnitude** of the **force** is normally written next to its arrow




EXAMPLES OF BASIC FORCE DIAGRAMS SHOWING DIRECTION AND MAGNITUDE. TWO OF THE FORCES IN THE SECOND DIAGRAM ARE UNKNOWN.

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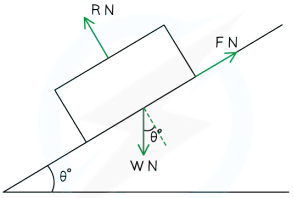
### What types of forces could be involved?

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

**friction (*F* N)** and **normal reaction (*R* N)**




SPECIFIC FORCES ACTING ON A BLOCK ON A ROUGH INCLINED PLANE

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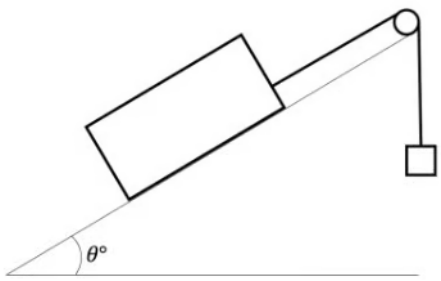
- Remember that the diagram is drawn to help understand the scenario - cars, blocks, etc are **modelled** as particles occupying a **single point** in space and so all **forces** acting on the car, block, etc act at that same **single point**
- The main forces that you will see are:
  - **tension** (a "pulling" force) acts away from a particle, **thrust** (a "pushing" force) acts towards it
  - **weight** is $W = mg$ where $m$ kg is the **mass** of the particle and $g$ is the **acceleration** due to **gravity** – usually $g = 9.8$ m s$^{-2}$
  - **friction** acts **parallel** to the surface in the **opposite** direction to **motion**
  - the **normal reaction** acts **perpendicular** to the surface (and **friction**)

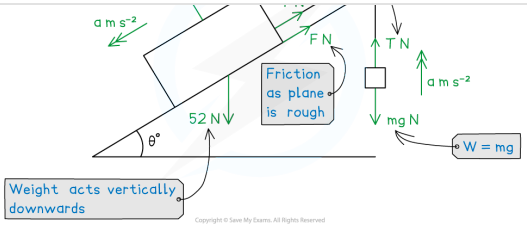
## Worked Example

The diagram below shows a block, of weight 52 N, at rest on a rough plane that is inclined at $\theta$° to the horizontal. The block is connected to a mass, $m$ kg, via a light inextensible string that passes over a smooth pulley. The particle hangs at rest vertically below the pulley.




Add all forces (known and unknown) to the diagram that are acting on the block and the mass. Also indicate the direction of acceleration of any subsequent motion. (You may assume the block is heavier than the mass and so any subsequent motion would see the block slide down the plane.)

**Answer:**




## Examiner Tips and Tricks

- Always draw a force diagram if appropriate.
- If a diagram is already given then add to it as you progress through the question.
- If a diagram is too small or it gets too complicated then draw a new diagram.
- You may be able to manage with just drawing the section of the diagram you are dealing with in any particular question part.

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